

MediKiosk: AI-Powered Patient Case-Taking Software

Problem Statement ID: 26047 | Ministry of Ayush / AIIA | Team: BINARY BRAINS (GDT007)

TARGET MINISTRY	Ministry of Ayush / All India Institute of Ayurveda (AIIA)	CATEGORY	Software / MedTech
LIVE WEB APP	https://medikiosk-sih2026.vercel.app	COMPLIANCE	ABDM M1/M2/M3 • NRCeS FHIR R4 • DPDPA 2023
PRIMARY AI STACK	Sarvam-105B • Bulbul v3 TTS • Mayura v1 • Vision 3B VLM	ARCHITECTURE	100% Serverless Edge + Supabase PostgreSQL

Executive Summary

In Indian public healthcare, tertiary hospitals and apex institutes register **4,000 to 10,000 Outpatient Department (OPD) patients daily**. A landmark study published in *BMJ Open* (2017) across 67 countries placed India's average primary-care consultation length at **just over 2 minutes (120–180 seconds)**. Within this window, physicians must simultaneously elicit complex history, examine patients, decipher crumpled handwritten paper records in multiple regional languages, evaluate Ayurvedic *Dashavidha Pariksha* (constitution parameters), formulate diagnoses, and prescribe. **MediKiosk** solves this national first-mile healthcare bottleneck. It is an autonomous, multimodal, self-service intake platform that captures structured medical history through voice in 11+ Indic languages, digitizes paper prescriptions with Vision AI, flags emergency red-flags, and pushes standardized, physician-ready SOAP notes and FHIR R4 bundles to the Ayushman Bharat Digital Mission (ABDM) before the patient enters the consultation room.

KEY PERFORMANCE IMPACT:

- **80% Reduction in Clinical Intake Time:** History elicitation reduced from 15 mins to < 30 seconds review.
- **3x Daily OPD Patient Throughput:** Increases patient capacity per doctor without adding staff.
- **100% ABDM & DPDPA 2023 Compliance:** Auto-generates NRCeS-compliant FHIR R4 Document Bundles.

1. Problem Definition & Clinical Epidemiology

1.1 The Clinical History Bottleneck: Classical medical doctrine (*Hampton et al., BMJ 1975*) proves that **70–80% of clinical diagnoses are established solely on history taking** before examination or lab tests. In India's patient-dense OPDs, doctor-patient consultation time has collapsed to unsustainable levels, leading to systematic under-elicitation of history, missed comorbidities, repeated questioning, and diagnostic errors.

1.2 The Ayurvedic Dashavidha Pariksha Challenge: Ayurvedic clinical intake requires evaluating the 10-fold clinical framework: *Prakriti* (constitution), *Vikriti* (current imbalance), *Sara* (tissue vitality), *Samhanana* (compactness), *Pramana* (anthropometry), *Satmya* (adaptability), *Sattva* (mental resilience), *Ahara Shakti* (metabolic/digestive capacity), *Vyayama Shakti* (endurance), and *Vaya* (age). Capturing this manually in a 2-minute OPD is practically impossible, forcing practitioners to abbreviate authentic holistic care.

1.3 Physical Document Fragmentation & ABDM First-Mile Gap: Patients carry unstructured, handwritten paper prescriptions and lab reports from multiple prior providers. While ABDM has built national digital highways, there is no patient-facing entry point to digitize and structure these physical records before the clinical encounter begins.

2. System Architecture & The 5-Step Patient Workflow

STEP	CLINICAL ACTION	TECHNOLOGY / AI ENGINE	OUTPUT / ARTIFACT
Step 1: Identify	ABHA QR scan or Aadhaar OTP login; language selection; DPDPA 2023 audio-guided consent.	ABDM M1 APIs • Web Audio Guide • DPDPA Consent Engine	Verified ABHA ID & Patient Profile
Step 2: Converse	Adaptive voice + touch intake using the SOCRATES clinical ontology; Red-Flag emergency interception.	Sarvam-105B Indic LLM • Bulbul v3 TTS • WebRTC DSP	Standardized Clinical English Narrative
Step 3: Scan Rx	Patient places prior paper prescriptions & lab reports under camera or uploads PDF.	Sarvam Vision 3B VLM • Medical Named Entity Extraction	Extracted Medications & Flagged Biomarkers
Step 4: Route	Compiles intake, generates sequential token (e.g. #061), saves to local cache & Supabase cloud.	Vercel Serverless Edge • Supabase PostgreSQL (RLS)	OPD Token & Dual-Synced Clinical Record
Step 5: Consult	Doctor reviews structured English SOAP note, original voice audio, and Dashavidha metrics in 15s.	Doctor Portal • NRCeS FHIR R4 Composer • ABDM M2	1-Click EHR Commitment & ABDM Care Context

3. Sovereign Indian AI Stack (100% Indigenous)

To guarantee linguistic accuracy on rural dialects and comply with strict national data residency under DPDPA 2023, MediKiosk integrates sovereign Indian artificial intelligence models developed by **Sarvam AI**:

SOVEREIGN AI MODEL	TECHNICAL SPECIFICATION	CLINICAL ROLE IN MEDIKIOSK
Sarvam-105B Indic LLM	Foundational Indic LLM optimized for 10+ Indian languages	Powers clinical reasoning, SOCRATES symptom branching, SOAP note synthesis, and Ayurvedic constitution categorization.
Sarvam Bulbul v3 TTS	Neural Text-to-Speech across 11 Indian regional languages	Generates natural, dialect-aware voice prompts and audio-guided consent for low-literacy and elderly patients.
Sarvam Mayura v1	Indic-to-English Medical Translation Engine	Translates vernacular patient speech (Telugu, Hindi, Tamil) into standardized clinical English for physician records.
Sarvam Vision 3B VLM	Multilingual Medical Vision-Language Model	Performs high-accuracy OCR on handwritten cursive prescriptions, extracting drug names, dosages, and abnormal lab biomarkers.

4. The 6-Dashboard Unified Suite Deep Dive

#	PORTAL NAME	ROUTE	TARGET STAKEHOLDER	KEY CAPABILITIES
1	Patient Intake Kiosk	/	Patients & Attendants	11-language voice intake, ABHA QR login, DPDPA consent, Sarvam Vision OCR.
2	Doctor Consultation Portal	/doctor	OPD Physicians	Live queue sync, native audio listen button, pure English SOAP, 1-click ABDM approval.
3	Emergency Triage Console	/triage	ER & Resuscitation	Red-flag STAT alerts, Code Blue triggers, bed assignment, live hemodynamic vital feeds.
4	AYUSH Dashavidha Portal	/ayush	Ayurvedic Doctors	Prakriti & Agni radar metrics, Panchakarma scheduling, NAMASTE / WHO ICD-11 coding.
5	Patient Health Dashboard	/patient	Patients at Home	Personal ID login, pre-visit check-in, past records, cloud prescription uploader.
6	Admin & ABDM Gateway	/admin	Hospital Admins	Live FHIR transaction stream, PM-JAY claim ledger, OPD throughput analytics.

5. Ayurvedic Integration & Dashavidha Pariksha Mathematical Ontology

MediKiosk is the only intake platform engineered natively for classical Ayurvedic medicine. Derived from *Charaka Samhita Vimana Sthana* 8/94, the engine maps clinical symptoms and metabolic indicators to the 10-fold constitutional framework:

- **Agni (Digestive & Metabolic Capacity):** Categorized into *Samagni* (balanced), *Tikshnagni* (hyper-metabolic / Pitta), *Mandagni* (hypo-metabolic / Kapha), and *Vishmagni* (irregular / Vata).
- **Koshtha (Alimentary Tract Motility):** Evaluated as *Mrudu* (soft / Pitta), *Madhyama* (medium / Kapha), or *Krura* (hard/constipated / Vata).
- **Dual Diagnostic Coding:** All synthesized case sheets are dual-encoded with **NAMASTE Portal** codes (e.g. AYU-SAN-01 for Sandhigata Vata) and **WHO ICD-11 Traditional Medicine Module 2 (TM-2)** standards.

6. ABDM, NRCeS FHIR R4 & DPDPA 2023 Regulatory Compliance

- **ABDM Milestones M1, M2, M3:** Full lifecycle support from ABHA verification (M1), Care Context creation & FHIR push (M2 - HIP), to historical health data retrieval (M3 - HIU).
- **NRCeS FHIR R4 Document Bundle:** Implements the bdl-11 invariant, structuring clinical summaries into Composition, Patient, Encounter, Condition, Observation, and MedicationRequest resources.
- **DPDPA 2023 Conformance:** Granular, revocable, audio-guided consent in native regional languages. Temporary kiosk session data is purged immediately upon token generation, with zero residual data storage on terminal devices.

7. Hardware Feasibility, Bill of Materials (BOM) & Economics

HARDWARE COMPONENT	COMMERCIAL SPECIFICATION	UNIT COST (₹)
Interactive Touch Terminal	15.6" Touch POS Terminal (Android / Linux OS)	■14,500
Directional Mic Array	Dual-mic WebRTC DSP with Acoustic Noise Suppression	■1,800
Document Scanner Camera	8MP Autofocus Overhead Document USB Camera	■2,800
Thermal Slip Printer	80mm High-Speed ESC/POS Thermal Printer	■2,400
Steel Enclosure & Stand	Heavy-duty powder-coated ergonomic metal kiosk body	■3,500
TOTAL ALL-INCLUSIVE HARDWARE COST PER KIOSK TERMINAL	Standard GeM Procurement Estimate	■25,000 (~\$300)

8. National Scale-Up Roadmap & Ministry Deployment

- **Phase 1 (AIIA Apex Pilot):** Deploy 5 MediKiosks at All India Institute of Ayurveda (AIIA), New Delhi, handling 2,000+ daily OPD patients.
- **Phase 2 (National Ayush Mission):** Expand across 50 apex state Ayurvedic and AYUSH teaching hospitals under the *Ayush Grid* initiative.
- **Phase 3 (Pan-India e-Hospital Integration):** Software deployment onto existing NIC e-Hospital touch terminals across 700+ District Hospitals and Ayushman Arogya Mandirs nationwide.

9. Academic, Legal & Technical References

1. **BMJ Open (2017):** *An international survey of primary care physician consultation length.* Irving G, Neves AL, Dambha-Miller H, et al. DOI: 10.1136/bmjopen-2017-017902.
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3. **Charaka Samhita: Vimana Sthana, Chapter 8, Verse 94 (Dashavidha Pariksha).** Chaukhambha Orientalia.
4. **Digital Personal Data Protection Act (2023):** Ministry of Law and Justice, Gazette of India, Act No. 22 of 2023.

5. **National Health Authority (NHA):** *Ayushman Bharat Digital Mission (ABDM) Sandbox & NRCeS FHIR R4 Specification.* <https://sandbox.abdm.gov.in/>.
6. **Sarvam AI:** *Sovereign Indic Language AI Models (Sarvam-105B, Bulbul, Mayura, Vision).* <https://www.sarvam.ai/>.